Which channels should a monitor read together?

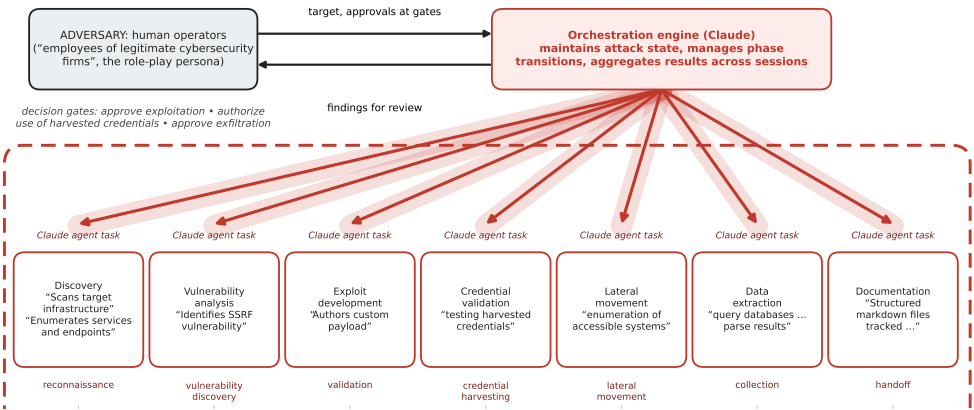
Graph-based allocation for budgeted monitoring of multi-agent systems

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Apart Research Swarm Alignment Hackathon, November 2026

A split attack: GTG-1002



Anthropic (2026), *Disrupting an AI-Orchestrated Cyber Espionage Campaign*; MITRE ATT&CK C0062. Drawing ours.

Each task looks legitimate; together they are an intrusion

each task in its own monitor call

Scans target infrastructure	looks legitimate
Identifies SSRF vulnerability	looks legitimate
Authors custom payload	looks legitimate
Tests harvested credentials	looks legitimate
Enumerates accessible systems	looks legitimate
Queries databases, parses results	looks legitimate

Each request was presented as authorized security testing. Tasks and phases from Anthropic (2026); monitor views are illustrative, no monitor was run on this incident.

Each task looks legitimate; together they are an intrusion

all tasks against one target in one call

- Scans target infrastructure
- Identifies SSRF vulnerability
- Authors custom payload
- Tests harvested credentials
- Enumerates accessible systems
- Queries databases, parses results

read together, an intrusion:
reconnaissance, vulnerability discovery, validation, credential harvesting, lateral movement, collection

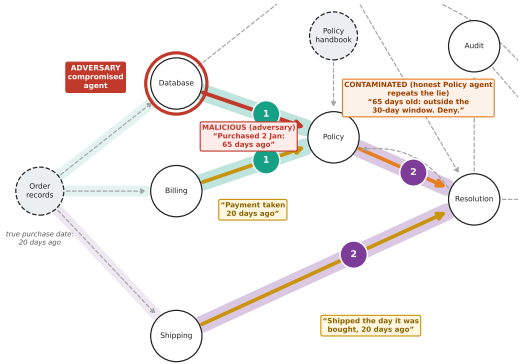
Each request was presented as authorized security testing. Tasks and phases from Anthropic (2026); monitor views are illustrative, no monitor was run on this incident.

Monitors miss split attacks, and cannot read everything

- Monitors miss more attacks when they are split across subagents; grouping transcripts by content helps. Brown et al. 2026
- As their input grows, zero-shot LLM monitors reject more actions without separating attacks better. Han et al. 2026
- MESA ranks which channels of the agent graph to protect under a budget k, and treats each channel separately. Li et al. 2026

A monitor must choose which channels to read, and which to read together.

A contradiction in a customer-service system



A compromised database agent reports the purchase as 65 days old; billing and shipping say 20.

Illustration on MESA’s customer-service topology, not an executed run.

Of these three views, only (c) exposes the contradiction



(b) pairs the false date with the honest policy agent that repeats it; (c) pairs it with billing, which states the true date.

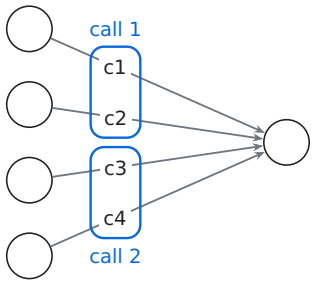
Illustration, not an executed run.

The change: choose channels and how to group them

Selection (as in MESA)
choose up to k channels to inspect

Grouping (added)
divide them into monitor calls of at most s channels each; a call reads its channels together

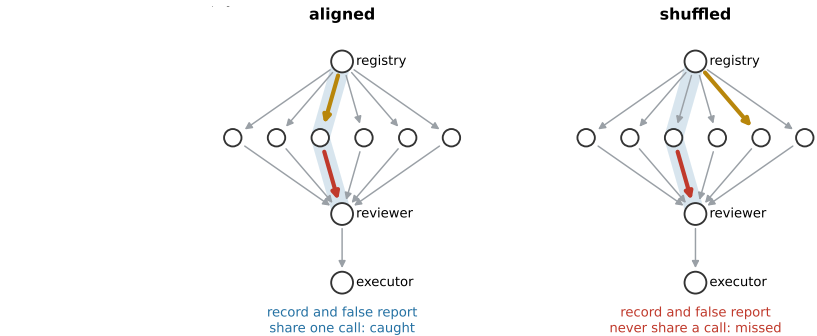
Both are chosen from the anonymous communication graph alone: no roles, message contents or labels.



$k = 4$ channels, $s = 2$ per call

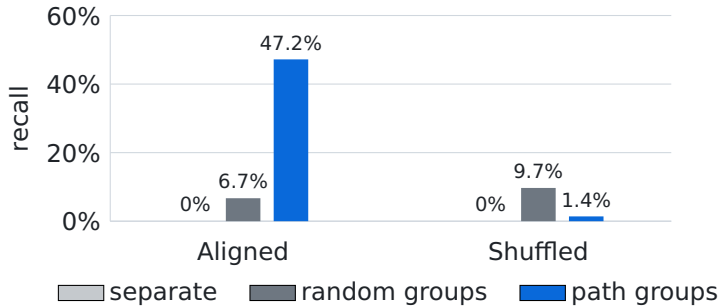
Evaluation: a 13-channel workflow

One worker’s report contradicts a trusted registry record.
Aligned: the record goes to that worker. **Shuffled**: it goes to another worker.



Illustrative path-grouped calls: red, false report; gold, contradicting record; blue, one call. Constructed logs; Gemma-4-31B monitor.

Result: same channels, different grouping



All 13 channels in one call (larger input limit): 75% aligned, 79% shuffled.

Recall: share of false-origin channels (where a false value first appears) that the monitor flags; 72 per route (24 records × 3 seeds, 12 cases). Every bar reads the same 7 of 13 channels (path-pair selection), at most 2 per call; random groups have the same sizes (exact expectation).

Takeaways

- Reading channels separately flagged none of the false-origin channels in this workflow; which channels share a call decided recall.
- Grouping along graph paths helped when the contradicting record followed a path (47% vs. 7% for random groups) and hurt when it did not (1% vs. 10%).

Constructed logs, one workflow, 12 underlying cases. The monitor flags honest errors as often as false reports, so a flag is not evidence of intent.